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Name:

Section:

**BRAC University**

Semester: Spring 2025

Course Code: CSE350

Digital Electronics and Pulse Techniques

Faculty: QSH

Set

A

Assessment: Quiz 3
Duration: 20 minutes
Full Marks: 20

Question 1

(a)	Determine the quantization levels for this Flash ADC.	[8]
(b)	Determine the full range ($V_{\max}$, $V_{\min}$) of the analog signal v_A .	[2]
(b)	What will be the binary output when $v_A = 3.4$ V? Elaborate on the full working process (which comparators give output HIGH and which encoder inputs are driven HIGH)	[10]

$$a) \quad v_0 = 1 \text{ V}$$

$$I = \frac{9-1}{5R} = \frac{1.6}{R}$$

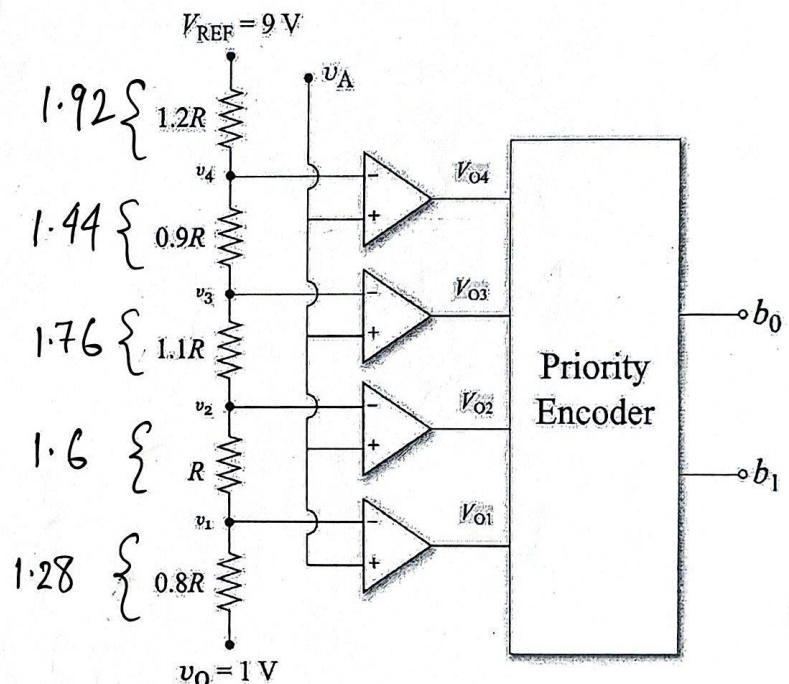
$$\therefore v_1 = 2.28 \text{ V}$$

$$v_2 = 3.88 \text{ V}$$

$$v_3 = 5.64 \text{ V}$$

$$v_4 = 7.08 \text{ V}$$

$$V_{\text{REF}} = 9 \text{ V}$$



$$b) \quad V_{\max} = 9 \text{ V}$$

$$V_{\min} = 1 \text{ V}$$

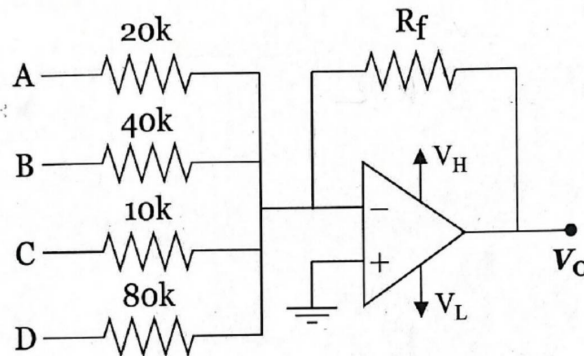
$$c) \quad v_A = 3.4 \text{ V falls between } v_1 \text{ and } v_2.$$

Thus, only v_{O1} will be High, rest will be Low.

Thus Binary output will be 00

(a) [20 marks] Consider the following DAC {For inputs, assume Binary 1 = -5V and Binary 0 = 0V}:

- What is the name of this particular DAC? [1 mark]
- Among the inputs (A, B, C, and D), determine MSB and LSB. What will be the output for input 1110? {Consider R_f to be 10k for this question only} [5 mark]
- Determine the value of R_f when the highest output is 7.5V. Also mention the values of V_H and V_L you need to set for the DAC to work properly. [14 mark]



i. Binary weighted DAC

ii. C is MSB } The order is C A B D
 D is LSB }

ii For 1110 $\Rightarrow \left(\frac{-5-0}{10} + \frac{-5-0}{20} + \frac{-5-0}{40} \right) (-10k\Omega)$

$$V_o = -\left(\frac{1}{2} + \frac{1}{4} + \frac{1}{8} \right) (-10k\Omega)$$

$$V_o = 8.75V$$

iii Highest output = 7.5

$$\therefore 7.5 = -\left(\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} \right) (-10k\Omega) R_f$$

$$\Rightarrow 7.5 = \frac{15}{16} R_f$$

$$\Rightarrow R_f = 7.5 \times \frac{16}{15}$$

$$= 8k\Omega$$